

Problem Statement — Retail Sales Analysis

Background

A retail superstore operates across multiple regions, selling products across three major categories: Furniture, Technology, and Office Supplies. While the business generates consistent revenue, leadership has noticed inconsistent profitability across product lines, regions, and customer segments — with a significant number of orders resulting in a loss.

Business Problem

The company needs a data-driven understanding of **where it is making money, where it is losing money, and why** — so that leadership can make informed decisions on pricing, discounting, inventory, and regional strategy.

Specifically, the business wants answers to:

- Which product categories and sub-categories are driving revenue vs. which are eating into profit?
- Which regions are underperforming, and is that tied to specific products or discount practices?
- Is there a relationship between the discounts offered and the profitability of an order?
- Who are the most valuable customers, and how can the business retain them?
- Are shipping practices (mode, delivery time) affecting cost and customer experience?
- What seasonal or monthly patterns exist in sales that could inform demand planning?

Objective

To analyze historical sales transaction data and deliver:

1. A cleaned, structured dataset ready for analysis
2. A set of SQL-based business insights answering the questions above
3. An interactive Power BI dashboard summarizing key metrics and trends
4. A short set of data-backed recommendations for improving profitability

Scope

Data: Historical order-level transaction data (Superstore Sales dataset) — includes order/ship dates, category, sub-category, product, customer, region, sales, profit, discount, quantity, and shipping details.

Tools used: Python (Pandas) for data cleaning and exploration, Microsoft SQL Server for business-question querying, Power BI for visualization and dashboarding.

Out of scope: Real-time data pipelines, external market data, and customer-level personal information beyond what's in the dataset.

Key Questions Analyzed

1. What is the total sales and profit by category?
2. Which sub-category is generating the most loss?
3. What is the sales and profit performance by region?
4. What does the month-wise sales trend look like (seasonality)?

5. What are the top 10 best-selling products (by sales and quantity)?
6. What are the top 10 products causing the most loss?
7. What is the relationship between discount and profit?
8. Which customers are the most valuable (top spenders)?
9. What is the average shipping/delivery time by ship mode?
10. What percentage of total orders result in a loss?
11. What is the month-over-month sales growth %?
12. How does ship mode affect sales and profit?

Expected Outcome

A clear picture of the business's profitability drivers, supported by a dashboard that lets stakeholders explore performance by category, region, and time period — along with actionable recommendations (e.g., revisiting discount policy on loss-making sub-categories, focusing retention efforts on top customers, or reviewing underperforming regions).